

AI Exposure at Three Levels: A Framework for Occupation, Task and Skill Measures of the Labour Market

Conway

Working paper. The framework and the measures are established; the test in Section 5 is in progress.

Contents

Extended summary

Introduction

Section 1: One wage bill written three ways

Section 2: Freund and Mann, and what an empirical counterpart needs

Section 3: Where each measure enters

Section 4: What the construction has established

Section 5: The test, and where it stands

Conclusion

References

Extended summary

Exposure to generative artificial intelligence has been measured at three levels of the labour market, and the levels are not interchangeable. Felten, Raj and Seamans (2021) score the fifty-two abilities of the Occupational Information Network, Eloundou, Manning, Mishkin and Rock (2023) score its task statements, and Srinivasan, Chen and Zakerinia (2024) score occupations, and applied work carries each to whatever level it needs through an aggregation that has not been examined. This paper sets out the framework in which the three levels are one wage bill written three ways, shows what each level can support, places each measure at the level where it was collected, and specifies the test that follows. The framework is built and its parts are checked, and the test that follows from it is half run, for a reason the framework itself supplies.

The framework is a nesting. The wage bill is the sum over occupations of employment times wage, an identity that holds whatever else is true. An occupation's wage is the sum over its tasks of a task share times a task price, and there are 17,383 task prices against 829 observed wages, so that description has more unknowns than facts and can never be contradicted. A task's price is the sum over skills of its skill content times a skill price, and there are twenty-five skill prices against the same 829 wages, so that description can fail, and it is the only one of the three that can. Which quantities in this literature are testable therefore follows from how many statements the Network's analysts wrote at each level, which is a property of a survey instrument. Freund and Mann (2026) build the same nesting in general equilibrium and project that the return to social skills rises and the return to analytical skills falls; they cluster tasks before estimating for the reason just given, and they do not test the projection against realised prices.

Each measure enters at one layer and reaches the others through the two matrices of the nesting, and the propagation can be checked where two measures meet. Carrying the Eloundou task ratings down to abilities reproduces the Felten ability scores at 0.667 for reading comprehension, 0.579 for translation and 0.546 for language modelling, against 0.289 for real-time video games and 0.172 for image recognition, so the propagation discriminates among AI applications in the right order and performs, on the applications that matter, at the 0.591 with which Eloundou's own human and machine raters agree. Inspected at their source, the capability measures are closer to one construct than their agreement suggests. The Srinivasan automation score is the Eloundou construction re-run on a later release with a later model, so their correlation of 0.833 shows reproducibility, and the Srinivasan augmentation score is one minus a Herfindahl index over the same classification, so a quadratic in automation accounts for 77 percent of it. Skill groups still explain 0.209 of what remains, with social skills, engineering and technology, arts and humanities, work output and psychomotor abilities on the augmenting side.

On the skill layer itself, twenty-five named groups from the Network's own content model explain occupational wages at a cross-validated fit of 0.648 in 2025, above the 0.544 of all 161 descriptors. Price levels are not recoverable, price changes are, and skill content explains 52 percent of the variation in occupational wage changes from 2022 to 2025. Employment weighting reduces 820 occupations to an effective 121, so inference comes from relabelling year transitions rather than from conventional standard errors, and four results in the course of this work reached conventional significance and did not survive that. Within an occupation, a task's rated exposure predicts the automation share of its observed use at a standardised value of 3.70, and only the machine rating predicts, which is a finding about the instruments before it is a finding about work.

The test has two halves. On the stock, a balanced panel of 693 occupations from 2012 to 2025, no skill group moved by more than 1.51 times its own year-to-year volatility after 2022, and every deviation from the pre-2020 path begins in 2020 or 2021, before language models were released. Realised wages and employment show nothing, and they cannot say whether that is because nothing has happened or because it has not yet reached the stock. The half that separates those readings is the flow: the task composition of job postings, which moves at monthly resolution across both the pandemic and the release of language models, and which no model has rated for relevance to AI. That half is specified here, with its groups, its pre-period and the reading of each possible outcome fixed before the data are seen.

Introduction

The empirical literature on generative artificial intelligence and work rests on a small number of exposure measures. They have been compared with one another and their disagreements noted, usually as a question of whether capability ratings or observed usage is the more informative. Less attention has gone to the fact that they were collected at different levels of description, so that comparing them at all requires a translation nobody has examined, and less still to the fact that the levels differ in what they can support.

Wages are observed once per occupation, and there are 829 of them in the 2025 Occupational Employment and Wage Statistics, carrying a wage bill of 10.70 trillion dollars. There are 18,796 task statements in the current Network release and 17,383 with the ratings this paper uses, so any set of task values has more unknowns than facts and cannot be contradicted by the data producing it. There are 161 descriptors, and after grouping twenty-five, so a set of skill values can fail. This paper takes that asymmetry as its organising fact. It writes the labour market as one wage bill at three levels, shows which level each existing measure was collected at and how a measure travels between levels, checks the travel where it can be checked, and states the test that the framework licenses. The framework is finished, and the test is not, because the data that can run its discriminating half are job postings rather than employment counts, and those are still being obtained.

Section 1 sets out the three layers as equations and counts what each can identify. Section 2 places Freund and Mann (2026), who build the same nesting in general equilibrium, and says what an empirical counterpart needs from them. Section 3 places each measure at its layer and reports what the measures turn out to be once inspected. Section 4 reports what the construction has established about the instrument and about the skill layer. Section 5 gives the test, the half of it already run, and the half specified for data still to come.

Section 1: One wage bill written three ways

An industry produces output, that output is produced by activities, and those activities draw on human capability, so occupations, tasks and skills describe one process at three distances. The premise runs from Autor, Levy and Murnane (2003) and Acemoglu and Autor (2011) through the skill literature descending from Katz and Murphy (1992). What follows is the arithmetic of that premise when the only prices the labour market reveals are occupational wages.

Let E_o be employment and w_o the mean wage in occupation o . The first layer is an identity:

$$Y = \sum_o E_o w_o.$$

It is a budget, so whatever is said at the other two layers has to add up to it, and because it holds by construction it can constrain a description without testing one. Quantities built as shares of Y return Y whatever the shares are, so a reconciliation to the national wage bill is a check that the arithmetic ran and nothing more.

The second layer opens the occupation. Occupation o performs tasks i in shares β_{oi} that sum to one, and each task carries a price p_i :

$$w_o = \sum_i \beta_{oi} p_i.$$

The shares come from the Network's task frequency ratings, published in seven bands from yearly or less to hourly or more and converted to an expected count per year. Frequency is

used because importance, the alternative, is compressed: in the median occupation the most important task rates 1.41 times the least important, against a median ratio of 14.78 on frequency, and a share built from importance says that every task in a job accounts for nearly the same part of the work. This layer is where the technology has been observed, since every measure of generative AI was built by judging a capability against a description of work. It is also the layer that cannot be tested, with 17,383 prices to recover from 829 wages.

The third layer opens the task. Task i draws on skills k with intensity γ_{ik} , and each skill carries a price π_k :

$$p_i = \sum_k \gamma_{ik} \pi_k, \quad \text{so that} \quad w_o = \sum_k s_{ok} \pi_k, \quad s_{ok} = \sum_i \beta_{oi} \gamma_{ik}.$$

The intensities come from the Network’s crosswalk from task statements to detailed work activities, which roll up to forty-one generalized activities and thence, through the published crosswalks from abilities, skills and knowledge to work activities, to the wider descriptor set. Built that way, 123 of the 161 descriptors vary across the tasks of an occupation. That step matters, because the cached parameter this work began from set every task’s skill mix to its occupation’s average, so that all 124,614 occupation by descriptor pairs containing more than one task carried a single value and the skill layer was a fixed linear image of one number per occupation. With the rebuilt γ_{ik} and twenty-five skill groups, the regression of w_o on s_{ok} asks twenty-five numbers to reproduce 829, and it can fail. It is the only one of the three layers where a claim can.

The skill grouping used throughout is the third level of the Network’s own content model, which gives twenty-five named groups across abilities, skills, knowledge and work activities. Figure 1 shows the three equations, the two matrices that connect them, and the layer at which each measure of exposure arrives.

Section 2: Freund and Mann, and what an empirical counterpart needs

Freund and Mann (2026) give the same nesting a general equilibrium form. Occupations bundle tasks, workers hold task-specific skills and sort by comparative advantage, and automating a task raises the weight on the tasks that remain, which is their mechanism of job transformation. Estimating the model on individual worker data, they project that the return to social and non-routine manual skills rises and the return to analytical skills falls. Their task weights come from language model estimates of time shares rather than from frequency bands, and their task-specific skill distributions are estimated rather than declared unrecoverable, so on two of the three layers their construction is the better one.

Two things about their paper are what this one is built around. They cluster roughly nineteen thousand task statements into thirty-eight categories before estimating, and that step is what makes their estimation feasible, which is the counting argument of Section 1 made in practice. Their projection is also only a projection, in that it is what the estimated model implies, and it

One wage bill written three ways, and the layer at which each measure arrives

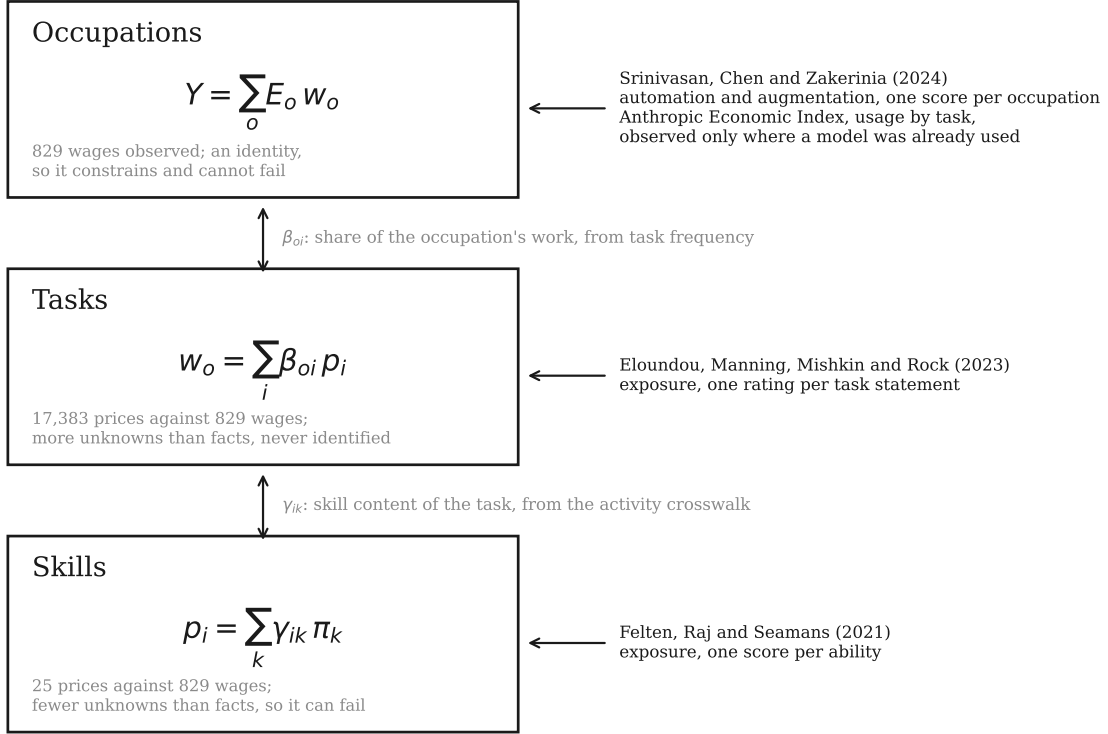


Figure 1: One wage bill written three ways, with each measure entering at the layer where it was collected. β_{oi} carries a task-level quantity to occupations and γ_{ik} carries it to skills; a quantity collected at occupations reaches skills only by inversion.

is never held against realised prices. An empirical counterpart has to state the projection in a vocabulary that wages can be regressed on, fix the membership of social and analytical before looking at any outcome, and run the test at the one layer where twenty-five numbers against 829 can be wrong. That is what the framework here supplies, and the test in Section 5 is written to their projection.

Section 3: Where each measure enters

Felten, Raj and Seamans (2021) enter at the skill layer. They surveyed respondents on the relatedness of ten AI applications to the fifty-two abilities of the Network, combined the resulting matrix with occupational ability requirements, and published the intermediate ability-level scores. Those abilities are fifty-two of the 161 descriptors used here, so the measure joins without propagation, all fifty-two after one renaming between releases. The ten applications come from 2016-era progress benchmarks and include image recognition, real-time video games and instrumental track recognition, so the headline index measures exposure to artificial intelligence broadly, and the application-by-ability matrix allows any one application to be isolated. For the present technology the language modelling and reading comprehension columns are the relevant ones.

Eloundou, Manning, Mishkin and Rock (2023) enter at the task layer. They rate 19,265 task statements on a four-point rubric separating tasks a language model can substantially accelerate from those it cannot, with separate ratings by human annotators and by GPT-4, of which 18,179 join the task set used here. Their headline index correlates 1.0000 with the machine labels and 0.591 with the human ones, so it is the machine rating, and Section 4 turns on that distinction. This is the measure Freund and Mann employ. It reaches occupations through β_{oi} and skills through γ_{ik} , and both directions are used below.

Srinivasan, Chen and Zakerinia (2024) enter at the occupation layer, scoring automation and augmentation for 911 occupations, of which 910 join the Network by name, giving 785 six-digit occupations and covering 681 of the 693 in the panel of Section 5. Both indices are built from the task descriptions of Network release 25.1 classified by GPT-4o on the Eloundou rubric. Job postings enter their paper to validate the indices and never to construct them, so neither score contains labour market outcomes. An occupation-level score reaches the skill layer only by inversion, regressing it on s_{ok} , and that inversion is reported below.

The Anthropic Economic Index also enters at the occupation layer, though it is built from tasks. It classifies conversations with a deployed model against Network tasks and reports the composition of interaction modes for each, its automation share being the sum of directive and feedback-loop modes and its augmentation share the sum of iteration, validation and learning, with a filtered residual completing one hundred percent, so the two correlate minus 0.588 by construction. It observes usage rather than capability, and it observes it only where a model was already used: coverage is 20.9 percent of the tasks considered here, the covered tasks are more exposed by 0.236 on a zero to one scale than those omitted, and the most used tenth accounts for three quarters of recorded usage.

The framework says where two measures should meet, and the first check is at abilities, which Felten measured directly and which the Eloundou ratings reach through γ_{ik} by a route sharing no data with the survey. Propagated exposure correlates 0.667 with reading comprehension, 0.579 with translation and 0.546 with language modelling, and 0.289 with real-time video games and 0.172 with image recognition. The ordering follows the closeness of each application to what a language model does, the video game column serves as a placebo, and since Eloundou’s human and GPT-4 raters agree on 65.6 percent of task labels with numeric scores correlating 0.591, the propagation performs at the reliability of its source on the applications that concern it. Variation survives the aggregation: averaging eighteen thousand task ratings into twenty-five groups gives a coefficient of variation of 0.292 across a range from 0.045 to 0.513, with mental processes highest, foundational literacy and numeracy close behind, and physical abilities lowest.

The second check is at occupations, where the Srinivasan automation score meets the Eloundou ratings carried up through β_{oi} . They correlate 0.833, and the reason is instructive. The Srinivasan score applies the Eloundou rubric to Network release 25.1 using GPT-4o, weighting the two exposed categories one and one half and weighting each task by its importance, so the correlation measures reproducibility across data vintages and model versions. Several capability measures in this literature share a rubric or a source of task descriptions, and agreement among them should be read accordingly.

Augmentation asks the same care. Srinivasan and co-authors define it as one minus a Herfindahl index over the shares of exposed and unexposed tasks in an occupation, on the reasoning that a job combining automatable work with work that resists automation gains most. That index is a quadratic in the exposed share, the automation score is a related summary of the same classification, and a quadratic in automation accounts for 77 percent of the variance in augmentation across occupations.

Inverting the two occupational scores onto the twenty-five skill groups needs no employment weighting, since exposure is not a share of the wage bill, so the effective sample stays at the number of occupations. Skill groups account for occupational exposure with cross-validated fit of 0.712 on automation and 0.415 on augmentation, and the named groups outperform ten principal components on both. On the raw scores, ten of twenty-five groups show an augmentation loading distinguishable from their automation loading, stable across five hundred bootstrap resamples. Removing the quadratic and inverting the residual leaves eleven groups distinguishable, with skill groups accounting for 0.209 of the residual variation, and the two profiles correlate 0.604. Social skills, engineering and technology, arts and humanities, work output and psychomotor abilities hold up; business and management falls from a statistic of minus 3.1 to plus 0.4 and mental processes from minus 2.9 to minus 1.2, so the apparent concentration of automation on cognitive and administrative work is largely the index restating an exposure share. Figure 2 shows both profiles.

Section 4: What the construction has established

The first set of results concerns the instrument. Importance in the Network is recorded on a five-point scale of which respondents use a narrow part, and converting such a rating into a share yields something close to uniform. A descriptor proximity matrix built from occupational similarity places 98.5 percent of its 798,342 pairs above 0.90, with a coefficient of variation of 0.022, and an occupation's profile across the forty-one generalized work activities sits a total variation distance of 0.10 from a uniform profile. The Network's two descriptions of what a job involves, the incumbent survey of work activities and the activity profile implied by the occupation's own tasks, agree at a median rank correlation of 0.33. Valuing 17,383 tasks by allocating each occupation's wage bill in proportion to task importance produces values correlating 0.997 on ranks with dividing that wage bill equally among the occupation's tasks, with eighty-seven of the largest hundred tasks identical under the two procedures, and substituting frequency lowers the correlation to 0.883. The Network is used because it is free, covers the whole economy, and supplies the vocabulary in which every measure in Section 3 is expressed, and these are the terms on which it does so.

The second set concerns the skill layer. Fitting occupational wages with descriptor intensities and evaluating on held-out occupations gives 0.554 for a single ordinal measure of required preparation, 0.700 for twenty principal components, 0.648 for the twenty-five named groups and 0.544 for all 161 descriptors, in 2025 and in the same order in every year from 2019. The market prices skills in something like twenty-five groups rather than 161, and the groups can

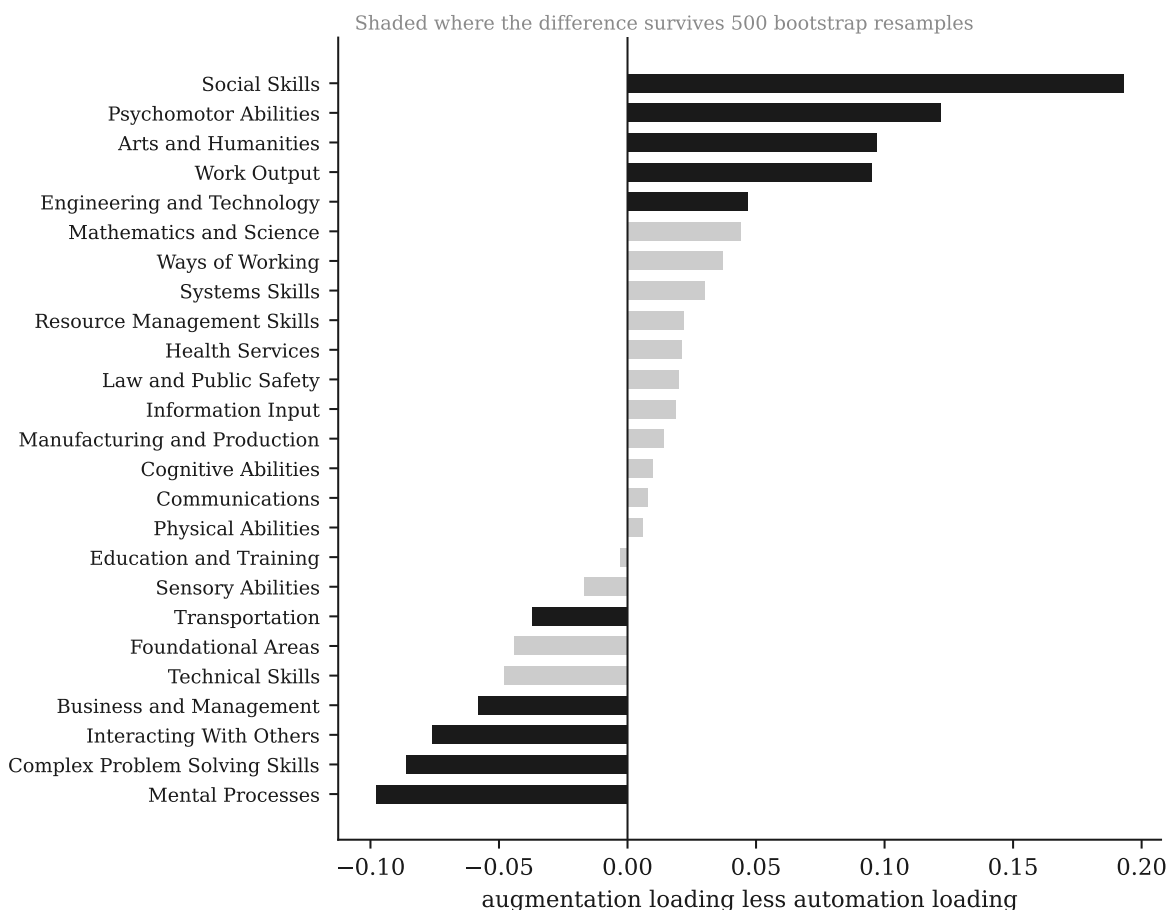


Figure 2: Augmentation loading less automation loading by skill group, from the inversion of the Srinivasan scores onto the twenty-five named groups. Shaded where the difference survives five hundred bootstrap resamples.

be the Network's own. Employment weighting, which a wage bill decomposition requires, gives an effective sample of 121 against 820 occupations in the sense of Kish (1965), the twenty largest occupations carrying a third of employment between them. Skill quantities, employment multiplied by intensity, involve no estimation and behave. Price levels are not recoverable: eleven of twenty-five groups take a negative price in some year, and constraining prices to be non-negative pins twenty of twenty-five at exactly zero. Price changes are recoverable, and skill content explains 52 percent of the variation in occupational wage changes between 2022 and 2025, so the test of Section 5 is run on changes.

Inference needs more care here than it usually receives. Comparing any two periods in these data returns between five and nine of the twenty-five groups apparently changing course when nothing has happened. Conventional standard errors are computed against the precision of a window's average rather than against the movement of the series between ordinary years, and composition series are smooth enough that a shift worth 0.17 percent of a level per year produces a statistic of 8.2. The procedure adopted throughout estimates the quantity for every year transition and builds the reference distribution by relabelling which transitions are treated, with the 286 relabellings of three among thirteen as the liberal reference and the eleven consecutive

triples as the conservative one. Four results in the course of this work reached conventional significance and did not survive it.

The last established result runs within occupations, and it is the one that shapes what Section 5 needs. The Anthropic index records only tasks a deployed model has already been used for, so asking whether exposure predicts usage would select on the outcome; asking whether exposure predicts the composition of that usage, conditional on the task being used at all, puts the selection on volume and the outcome on mode. On 1,617 tasks across 349 occupations, with occupation means removed, a task's Eloundou exposure predicts the automation share of its observed use at a slope of 0.109 and a standardised value of 3.70 against two thousand within-occupation shuffles, survives controlling for usage, and holds in three of four index releases. Substituting the human rating of the same tasks gives a slope of minus 0.008 and a standardised value of minus 0.38, too far from the machine result to be attenuation, since the 0.356 correlation between the two ratings would leave a slope near 0.039 if the human rating were the same signal with more noise; and on the 747 tasks where the ratings differ by more than 0.4, entering both gives 0.204 to the machine and 0.029 to the human. Both sides of that relationship are language model outputs. These data cannot distinguish the machine being the more accurate rater from the two sharing method variance, and separating those readings requires a task-level outcome that no model has rated for relevance to AI.

Section 5: The test, and where it stands

[WORK IN PROGRESS]

The test is the Freund and Mann projection, stated at the skill layer, on changes, with membership fixed from their vocabulary before any outcome was examined. Social is the Network's social skills and interacting with others groups; analytical is its foundational areas, complex problem solving, systems skills, mental processes and mathematics and science groups; manual is its physical abilities, psychomotor abilities and work output groups, which they project to rise with social. The quantity under test is the change in what each group earns, relative to its pre-period path, after the release of language models in November 2022, and the projection is that social rises against analytical.

It has two halves, one on the stock of employment and one on the flow of hiring, and they answer different questions, of which the first has been put and answered. The panel covers 693 occupations from 2012 to 2025, with a wage bill rising from 5.15 to 8.95 trillion dollars, assembled from the Occupational Employment and Wage Statistics and mapped across the 2018 revision of the occupational classification using the Network's crosswalk, which raises coverage of the 2018 wage bill from 0.813 to 0.936. Across 2022 to 2025 no skill group moved by more than 1.51 times its own year-to-year volatility. Five of twenty-five exceed two of their own standard errors, against a median of four across the 286 relabellings, and quantities give a probability of 0.455 weighted by headcount. The directional test is not supported on wages and runs the other way, the returns to both social and analytical content having risen against their pre-period paths with analytical rising further; on employment the social against analytical contrast moves as

projected, at a probability of 0.021 against all relabellings and 0.273 against consecutive triples, on the one specification carrying no pre-existing trend. Figure 3 shows the deviations from the pre-2020 path by year. On wages the residual runs at minus 0.005 in 2020 and reaches minus 0.025 by 2025; on employment the series takes an excursion of minus 0.040 in 2020 and stays positive from 2021. Both breaks precede the technology by two years, and the pandemic moves these series by more than the effect sought.

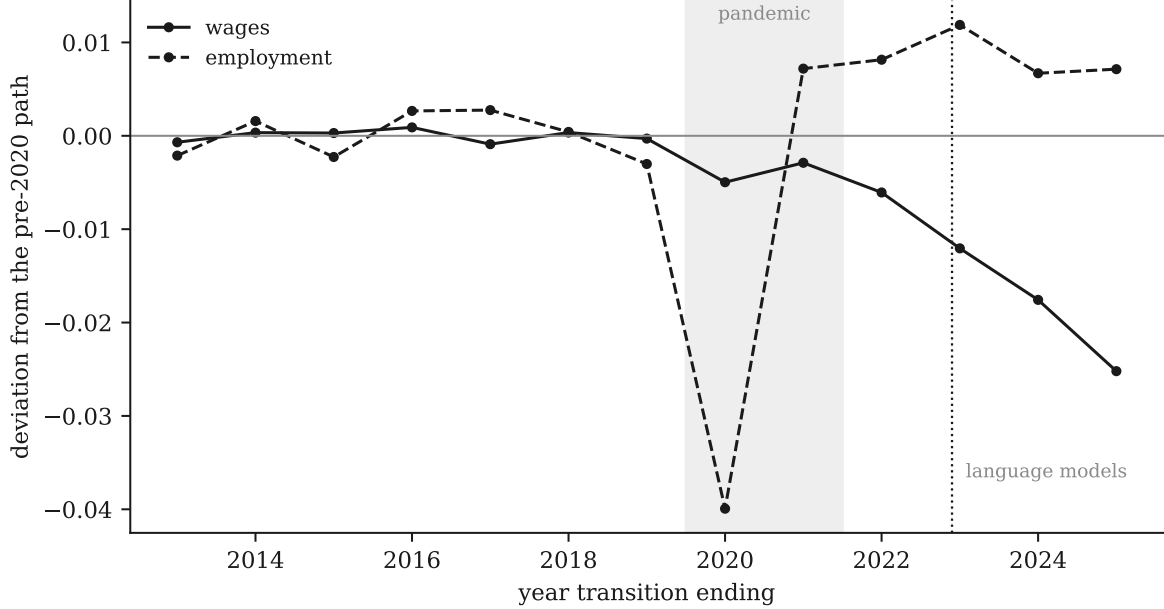


Figure 3: Deviations from the pre-2020 path by year, on wages and on employment. The pandemic is shaded and the release of language models marked. Every break begins before the technology.

The stock half therefore returns a null and cannot say what the null means. Employment is a stock, it moves through hiring frictions, notice periods and sticky wages, and a change in what employers want reaches it years after it reaches what employers ask for. A null on the stock through 2025 is consistent with nothing having happened and with something having happened that has not yet arrived, and annual data straddling a pandemic cannot separate the two, since the only break they contain is the wrong one.

The flow half is what separates them, and it is specified here for data still to be obtained. Meisenbacher, Nestorov and Norlander (2025) map more than 155 million online job postings from the National Labor Exchange to Network task identifiers, monthly from 2015 to 2025, and publish the toolkit that does so. Three properties of that series make it the right second half. It is monthly, so March 2020 and November 2022 are thirty-two observations apart rather than adjacent annual points, and the pre-period trend can be fitted and the two breaks read separately. Its outcome is whether a task appears in a posting, which no model has rated for relevance to AI, so the shared-method problem that ends Section 4 does not arise. And it observes intent rather than realisation: a posting is a fresh decision about what an employer wants next, so if the technology is changing work it shows there first.

The design is the within-occupation comparison of Section 4 run on a different outcome. For each occupation and month, the share of postings carrying each of its tasks; for each task, its Eloundou exposure; the question, whether tasks rated more exposed have been leaving postings faster since November 2022 than their pre-2020 trend and their occupation's other tasks would predict, with inference by relabelling months as before. Task composition is the outcome rather than any mention of artificial intelligence in the posting, because mentions follow fashion and 2023 and 2024 were years of it, and an occupation that a language model rates as exposed is also an occupation whose postings are likely to name the technology whether or not the work has changed. Whether a task quietly leaves job descriptions is a harder thing to imitate, and the boilerplate that job descriptions carry biases that outcome towards finding nothing rather than towards finding something spurious.

Each outcome has a reading fixed in advance, so that whatever the postings show has already been assigned its meaning. If exposed tasks are leaving postings within occupation and the stock has not moved, the null of the stock half means too early, and the size of the gap between intent and realisation is the result. If neither has moved, the null stands on the fastest-moving series available and is worth far more than it is at present. If the stock moves and postings do not, the mechanism is something other than job transformation, which is the mechanism Freund and Mann's model turns on. The postings cover the jobs that get advertised, which skews towards larger employers, white-collar work and external recruitment, so any result is about demand at that margin; the exchange upgraded its data warehouse in 2021 and the monthly counts before that are uneven, which any pre-period fit has to allow for; and the postings themselves are increasingly written with language models, which the authors read in the rising readability of the text, so the boilerplate is itself changing. Access to the underlying series is through the exchange's Research Hub, and that application is where this paper currently stands.

Conclusion

Three literatures measure exposure to generative artificial intelligence at three levels of one labour market, and this paper writes those levels as one wage bill three times over, shows that only the skill layer can be tested, places each measure at the layer where it was collected, and checks the passage between layers where two measures meet. The passage from tasks to abilities discriminates among AI applications in the right order. The capability measures are closer to a single construct than their agreement suggests, with augmentation largely a transformation of automation and a residual that still carries skill structure. The skill layer prices in about twenty-five groups, its price changes are recoverable and its levels are not, and its inference has to be built from the movement of the series between ordinary years. Within occupations a relationship appears, and it holds only for the machine-rated exposure, which is why the test that remains needs an outcome no model has rated.

That test is the Freund and Mann projection run against realised prices, and it is half done. The stock shows nothing through 2025, with every break preceding the technology, and cannot say whether nothing has happened or nothing has yet arrived. The flow can, and it is specified above

with its groups, its pre-period and its readings fixed. The framework says where the answer can be found, and obtaining the series that carries it is the work now under way.

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